

AMAL A K

Full-Stack Software Engineer · MERN · PERN · SaaS · TypeScript · AI Integration
Trivandrum, Kerala | +91 9605655082 | amal.anandvijay@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Full-Stack Software Engineer with hands-on experience architecting and delivering scalable multi-tenant SaaS applications using MERN and PERN stacks. Built DiagnomiQ, a production-grade healthcare SaaS platform featuring Database-per-Tenant isolation, automated Cloudflare DNS provisioning, JWT-based RBAC, AWS S3 data isolation, and Amazon Bedrock RAG integration. Proven in owning features end-to-end across healthcare, e-commerce, and edtech domains.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript

Frontend: React.js, Next.js, Redux, Redux Toolkit, Tailwind CSS, Shadcn/UI, Material UI, Recharts, Vite

Backend: Node.js, Express.js, RESTful APIs, JWT Authentication, RBAC, Middleware Architecture

Databases & ORMs: PostgreSQL, MongoDB, Prisma ORM, Sequelize, Mongoose, NeonDB

Cloud & DevOps: AWS S3, Amazon Bedrock (RAG), Cloudflare DNS, Vercel, Render, Netlify, Docker (basic), CI/CD

Architecture: Multi-Tenant SaaS, System Design, Microservices, Performance Optimisation, Agile/Scrum, GitHub

PROFESSIONAL EXPERIENCE

Software Engineer | Softnotions Technologies, Technopark, Trivandrum

Jan 2026 – Apr 2026

- Architected and led full-stack delivery of DiagnomiQ, a production-grade multi-tenant healthcare SaaS built on PERN stack (PostgreSQL, Express.js, React, Node.js), successfully migrating legacy clinical systems to a scalable cloud platform.
- Designed Database-per-Tenant architecture using PostgreSQL + Sequelize ORM, achieving 100% tenant data isolation across dynamically provisioned databases — meeting strict SaaS security and compliance requirements.
- Built fully automated tenant onboarding pipeline including dynamic PostgreSQL database creation, Cloudflare DNS subdomain provisioning per tenant, and AWS S3 folder-level data isolation — reducing manual setup time from days to minutes.
- Engineered JWT-based authentication with Role-Based Access Control (RBAC) and custom Tenant DB Middleware for secure, dynamic request routing across isolated per-tenant databases.
- Architected the central Platform Database serving as the SaaS control plane, managing tenant registrations, subscription plans, and all system-wide operations.
- Integrated Amazon Bedrock RAG pipeline — developed structured medical knowledge base and connected AI model via Bedrock API to deliver real-time clinical decision support in the live platform.

MERN Stack Developer | Relaxplzz Technologies, Technopark, Trivandrum

Apr 2025 – Aug 2025

- Developed scalable frontend features and RESTful APIs for GiftoMart, a production e-commerce platform using React.js, TypeScript, Tailwind CSS, MongoDB, Node.js, and Express.js.
- Built reusable TypeScript component library that improved code maintainability and accelerated cross-team development; collaborated effectively with product, design, and QA teams in Agile sprints.

MERN Stack Developer | SciPy Technologies, Trivandrum

Jun 2023 – Jan 2024

- Developed RESTful APIs in Node.js and responsive React UIs with Redux state management for a microservices-based ERP system serving an educational institution.
- Managed MongoDB storage with Mongoose schema design, integrated third-party APIs via Axios, and validated endpoint contracts with Postman across Agile sprints.

FEATURED PROJECTS

DiagnomiQ - Multi-Tenant Healthcare SaaS | React · TypeScript · Node.js · PostgreSQL · Sequelize · AWS S3 · Cloudflare · JWT · Amazon Bedrock

- Database-per-Tenant architecture with automated PostgreSQL provisioning, Cloudflare subdomain creation per tenant, and AWS S3 folder isolation for HIPAA-aligned medical data separation.
- RAG pipeline on Amazon Bedrock — structured medical knowledge base with Bedrock API integration for AI-powered clinical decision support built into the live SaaS platform.
- Built oncology risk dashboards using Recharts (IPI and aalPI survival probability charts) providing oncologists real-time visual risk stratification for lymphoma patient care.

FinSight — Personal Finance Tracker | Next.js · TypeScript · PostgreSQL · Prisma ORM · Tailwind CSS · Shadcn/UI · Vercel

- Built a type-safe full-stack finance tracker using Next.js, Prisma ORM, and PostgreSQL, featuring interactive analytics dashboards with category-based filtering — deployed on Vercel with edge optimisation.

GiftoMart — Gift Management Platform | React · Node.js · Express.js · MongoDB · TypeScript · Mongoose

- Developed gift creation and order-tracking modules with JWT authentication, reusable TypeScript components, and RESTful APIs using Node.js, Express.js, MongoDB, and Mongoose.

EDUCATION

B.Sc. Computer Science | Emmanuel College, Vazhichal

2018 – 2021